



CCP6224 Object-Oriented Analysis and Design

Trimester October/November 2025 – T2530

Lab Exercise

(Seminar Management System)

Group No. TC2L TT8L Group 7

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Table of Contents

1. INTRODUCTION	3
1.1 Background of Study	3
1.2 Problem Statement	3
1.3 Project Objectives	4
1.4 Project Scope	4
1.5 Significance of the Project	4
2. UML CLASS DIAGRAM	5
3. SEQUENCE DIAGRAM	6
3.1 Student Diagram	6
3.2 Evaluator Diagram	6
3.3 Coordinator Diagram	7
4. USER INTERFACE	8
4.1 STUDENT UI	8
Login Page	8
Student Registration Page	8
Student Main Page	9
Upload New File	10
Edit Project Details	11
4.2 EVALUATOR UI	12
Evaluator Main Page	12
Evaluator Session Page	12
Evaluator Select Session Due Date and Time	13
Grade with Marking Rubric and Remark	13
View All Files uploaded by Students	14
4.3 COORDINATOR UI	15
Coordinator Main Page	15
Create Session	15
Add New Evaluator	16
Session Details	16
Assign Session for Students	17
Generate Report	17
Generate Text File (txt) by System	17
Generate Excel File (CSV) by System	19

1. INTRODUCTION

1.1 Background of Study

In the landscape of higher education, academic seminars serve as a crucial platform for students to present their research findings, demonstrate technical communication skills, and receive constructive feedback from faculty members. As the student population grows, the administrative burden of organizing these events increases exponentially. Traditionally, the management of seminar sessions—ranging from scheduling and evaluator assignment to the collection of presentation materials—has been conducted through manual processes involving physical forms, disparate spreadsheets, and email communications.

While these traditional methods are functional, they are increasingly becoming obsolete in a digital-first educational environment. The reliance on manual data entry often leads to data redundancy, scheduling conflicts, and delays in grade dissemination. To address these challenges, this project introduces the **Seminar Management System (SMS)**. Developed using the Java programming language and Swing for the Graphical User Interface (GUI), this system aims to centralize and automate the entire lifecycle of seminar management, ensuring a seamless experience for coordinators, evaluators, and students alike.

1.2 Problem Statement

Despite the importance of seminars, the current manual workflow suffers from several critical inefficiencies that this project aims to resolve:

1. **Inefficiency in Reporting and Data Management:** Coordinators often spend excessive hours manually compiling scores and generating status reports. There is a lack of automated tools to instantly generate finalized schedules or evaluation summaries in standard formats like Excel (CSV) or text files.
2. **Human Error in Grading:** The manual calculation of marks based on specific rubrics (e.g., Presentation Flow, Content, Q&A) is prone to arithmetic errors, potentially affecting the fairness of academic assessment.
3. **Fragmented File Submission Process:** Students typically submit presentation slides via thumb drives or emails, leading to version control issues or lost files. Evaluators often face difficulties in retrieving the correct documents during the presentation session.
4. **Lack of Real-Time Information:** Students often lack a centralized dashboard to view their specific session details, assigned evaluators, and submission deadlines, leading to miscommunication and missed due dates.

1.3 Project Objectives

The primary objective of this project is to develop a robust, object-oriented Seminar Management System. The specific objectives are as follows:

- **To develop a role-based centralized platform** that distinguishes between Coordinators, Evaluators, and Students, providing specific functionalities tailored to each user's responsibilities.
- **To automate the evaluation and grading process** by implementing a digital rubric interface that automatically calculates total scores and stores qualitative feedback, eliminating calculation errors.
- **To streamline file management** by creating a secure, local storage mechanism that allows students to upload files with automated naming conventions (e.g., ID_Name_Date), ensuring easy retrieval by evaluators.
- **To facilitate automated reporting** by integrating a ReportGenerator module capable of producing formatted Seminar Schedules, Evaluation Reports, and Award Lists, with support for exporting data to CSV format for external analysis.

1.4 Project Scope

The scope of the Seminar Management System is defined by the functionalities available to its three core user roles:

1. **The Coordinator:** Acts as the system administrator with the authority to create and delete presentation sessions, manage user registrations, and generate comprehensive analytical reports. The coordinator can view the overall status of all sessions and export "Final Evaluation Reports" to Excel-compatible formats.
2. **The Evaluator:** Faculty members responsible for assessing students. The system allows them to view their assigned sessions, open student-uploaded documents directly from the dashboard, and input grades via a structured rubric table.
3. **The Student:** The end-users who can view their assigned presentation time, venue, and evaluator. They are provided with features to upload presentation materials and view their final results and feedback once the evaluation is complete.

1.5 Significance of the Project

The implementation of this system offers significant benefits to the academic administration. By transitioning from a manual to a computerized system, the university can achieve a "paperless" workflow, reducing environmental impact. For the faculty, the automated score calculation and centralized file access drastically reduce the time spent on administrative tasks, allowing them to focus on the quality of assessment. Finally, for students, the system guarantees transparency and clarity regarding their schedules and assessment criteria, contributing to a more organized and stress-free academic experience.

```

classDiagram
    class User {
        -userID: String
        -username: String
        -password: String
        -role: enum(student, evaluator, coordinator)
        +register()
        +login()
        +getId()
        +getName()
        +getPassword()
    }
    class SeminarMgtSystem {
        + studentList[]
        + coordinatorList[]
        + evaluatorList[]
        + sessionList[]
        + login(id, pass): boolean
        + assignParticipants(sessionID)
        + processEvaluations()
        + exportSummaryReport()
    }
    class PresentationType {
        <<enumeration>>
        oral
        poster
    }
    class Student {
        - studentID : String
        - researchTitle: String
        - projectAbstract: String
        - remark : String
        - sessionID : String
        - supervisorName: String
        - score : int
        - presentationType: PresentationType
        - wonAward
        - awardType
        + addSubmission(): String
        + uploadMaterial(filePath: string): Void
        + viewEvaluation(): Void
    }
    class Evaluator {
        - evaluatorID: String
        - evaluatorName: String
        - rubricCriteria[]
        - maxRubricScore = 25
        + viewSubmissions(sessionID: String)
        + evaluateSubmission()
        + filterBySession(sessionID: String)
        + getMyEvaluations()
    }
    class Coordinator {
        - coordinatorID: String
        - coordinatorName: String
        + createSession(): Session
        + assignEvaluator()
        + assignPresenter()
        + generateSchedule()
        + generateEvaluationReport()
        + generateAwardReport(): Report
        + computeAwardWinners(): void
    }
    class Submission {
        - submissionID: String
        - savedFileName: String
        - uploadTime: Date
        - status: submissionStatus
        + getSavedFileName(): Void
        + updateFile(filePath: string): Void
    }
    class Evaluation {
        - submission: Submission
        - evaluator: Evaluator
        - problemClarity: String
        - methodology: int
        - results: int
        - comments: String
        + getTotal(): Double
    }
    class Award {
        - awardID: String
        - awardType: String
        - criteria: String
        - winnerID: String
        + calculateWinner(): Student
        + assignWinner(student: Student): void
        + getAwardDetails(): Award
    }
    class Session {
        - sessionID: String
        - sessionDate: Date
        - sessionVenue: String
        - sessionType: String
        - sessionTime: Date/Time
        - evaluatorName
        - capacity: int
        - submissionDueDate: String
        - boardID: String
        + getSessionID()
        + getDate()
        + getTime()
        + getVenue()
        + getEvaluatorName()
        + getCapacity()
        + getSubmissionDueDate
        + getSessionType()
        + getBoardID()
    }
    class DataStore {
        - users: List<User>
        - submissions: List<Submission>
        - evaluations: List<Evaluation>
        - sessions: List<Session>
        - posterBoards: List<PosterBoard>
        - awards: List<Award>
        + evalsFor(submissionID): List<Evaluation>
        + computeAwards(): void
        + avgScore(submissionID): double
    }
    class Report {
        - reportID: String
        - content: String
        - generatedDate: Date
        + generateSeminarSchedule()
        + generateEvaluationReport()
        + generateAwardReport()
        + computeAwardWinners()
    }

    User "1" -- "*" SeminarMgtSystem
    User "1" -- "*" Student
    User "1" -- "*" Evaluator
    User "1" -- "*" Coordinator
    User "1" -- "*" Submission
    User "1" -- "*" Evaluation
    User "1" -- "*" Award
    User "1" -- "*" Session
    User "1" -- "*" DataStore
    User "1" -- "*" Report

    Student "1" -- "*" Submission
    Student "1" -- "*" Evaluation
    Student "1" -- "*" Award
    Student "1" -- "*" Session
    Student "1" -- "*" DataStore
    Student "1" -- "*" Report

    Evaluator "1" -- "*" Submission
    Evaluator "1" -- "*" Evaluation
    Evaluator "1" -- "*" Award
    Evaluator "1" -- "*" Session
    Evaluator "1" -- "*" DataStore
    Evaluator "1" -- "*" Report

    Coordinator "1" -- "*" Submission
    Coordinator "1" -- "*" Evaluation
    Coordinator "1" -- "*" Award
    Coordinator "1" -- "*" Session
    Coordinator "1" -- "*" DataStore
    Coordinator "1" -- "*" Report

    Submission "1" -- "*" Evaluation
    Submission "1" -- "*" Award
    Submission "1" -- "*" Session
    Submission "1" -- "*" DataStore
    Submission "1" -- "*" Report

    Evaluation "1" -- "*" Award
    Evaluation "1" -- "*" Session
    Evaluation "1" -- "*" DataStore
    Evaluation "1" -- "*" Report

    Award "1" -- "*" Session
    Award "1" -- "*" DataStore
    Award "1" -- "*" Report

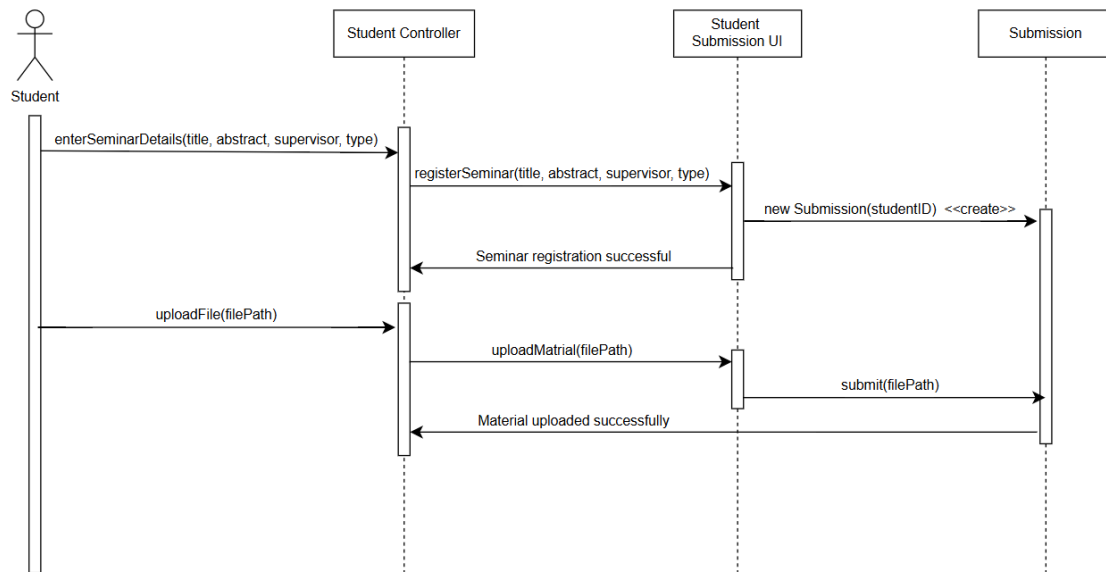
    Session "1" -- "*" DataStore
    Session "1" -- "*" Report

    DataStore "1" -- "*" Report

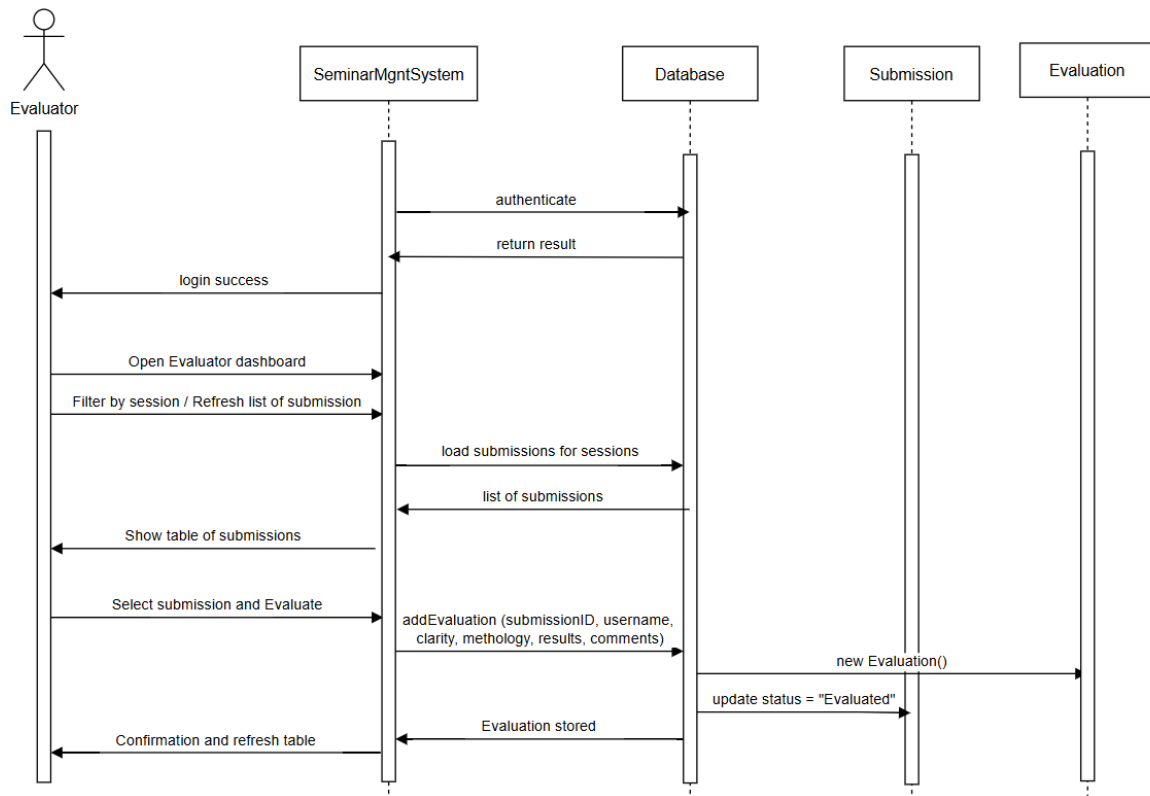
```

3. SEQUENCE DIAGRAM

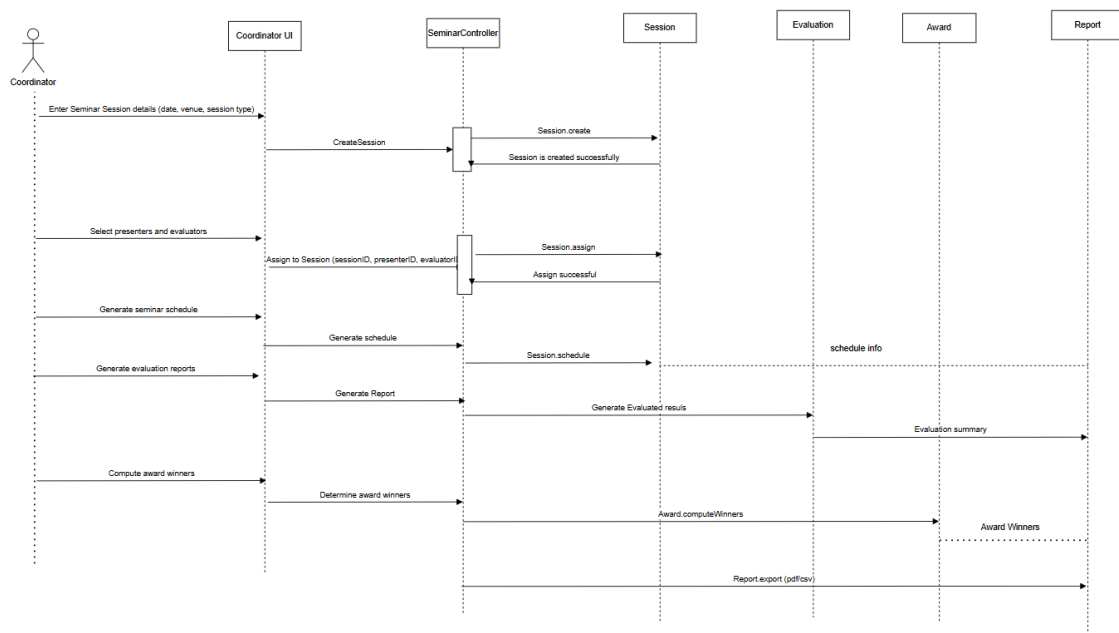
3.1 Student Diagram



3.2 Evaluator Diagram



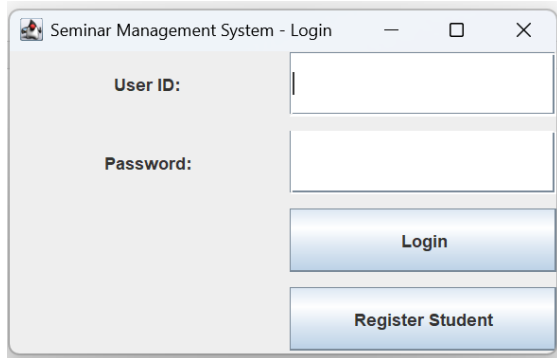
3.3 Coordinator Diagram



4. USER INTERFACE

4.1 STUDENT UI

Login Page



Seminar Management System - Login

User ID:

Password:

Login

Register Student

Student Account: (ID / PASSWORD)

Alice: S101 / 1

Bob: S102 / 1

Charlie: S103 / 1

Evaluator Account: (ID / PASSWORD)

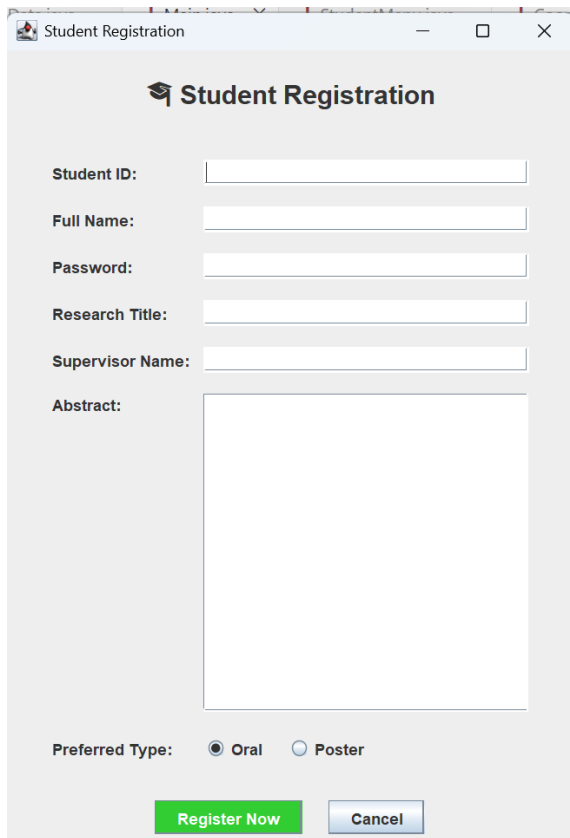
Dr. Tan: E001 / 1

Coordinator Account: (ID / PASSWORD)

Dr. Admin: C001 / 1

Student Registration Page

Click the “Register Student” button to register Student Account



Student Registration

Student ID:

Full Name:

Password:

Research Title:

Supervisor Name:

Abstract:

Preferred Type: ☒ Oral ☐ Poster

Register Now Cancel

Student Main Page

After login successful, students able to view the main page. It shows Session Details, Project information, and Submission details in main page.

Student Dashboard - Alice

Welcome, Alice (S101) [Logout →](#)

Session: SES-01

Evaluator: Dr. Tan (E001)

Presentation Time: 2026-02-10 10:00 AM

Venue: Lab 1

My Project Details

Title: AI Research

Supervisor: Dr. Lee

Presentation Type: Oral

Abstract:

Uploaded Files (Double-click to open):

File Name	Upload Time

Submission Due Date: 2026-02-10 11:59 PM

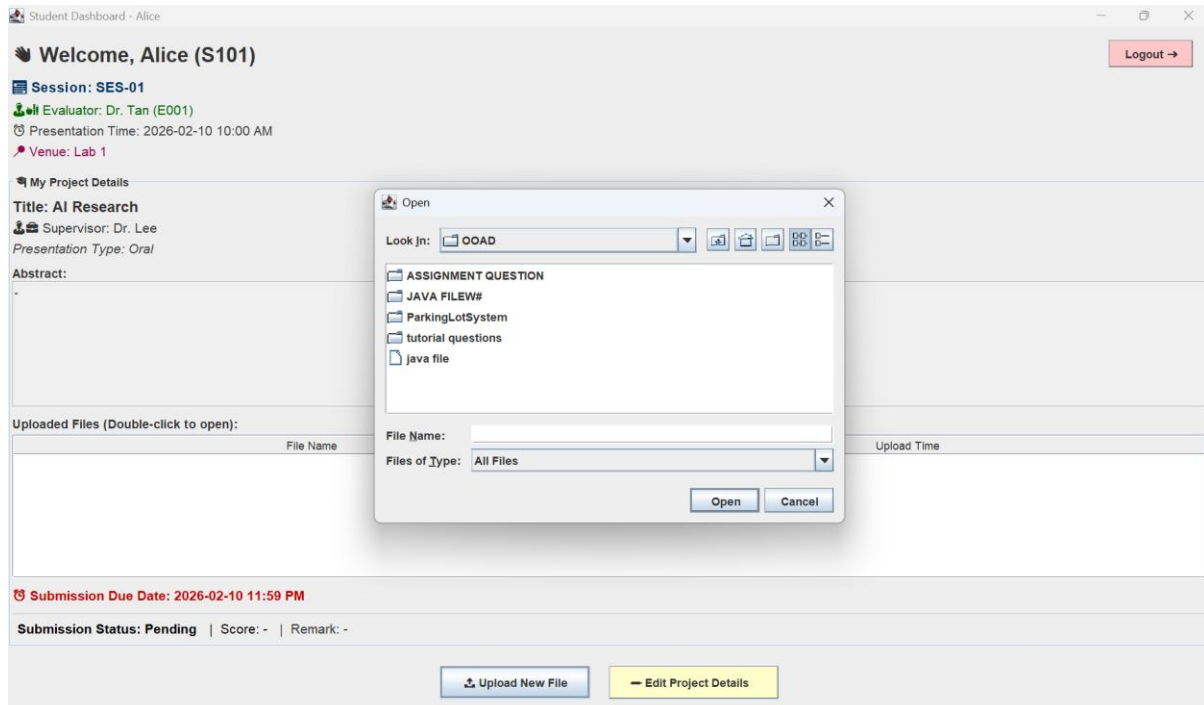
Submission Status: Pending | Score: - | Remark: -

[Upload New File](#) [Edit Project Details](#)

Upload New File

Students can upload and submit any Files into this System.

They also can double click Upload Files in the main page to view all the files they uploaded.



Edit Project Details

Students able to edit the Project Details when they click the button call “Edit Project Details”.

They can edit:

1. Research Title
2. Supervisor
3. Presentation Type
4. Project Abstract

The screenshot displays a web application interface for a student dashboard. The main content area shows a welcome message for Alice (S101) and session information (SES-01). A modal window titled "Edit Project Details" is open, allowing the user to update their project information. The modal contains the following fields and options:

- Research Title:** A text input field containing "AI Research".
- Supervisor:** A text input field containing "Dr. Lee".
- Presentation Type:** Two radio button options: "Oral Presentation" (selected) and "Poster Presentation".
- Project Abstract:** A large text area for entering the project abstract.

At the bottom right of the modal is a green "Save Changes" button. The background dashboard includes a sidebar with "My Project Details" showing the current project title, supervisor, and presentation type. At the bottom of the dashboard, there are buttons for "Upload New File" and "Edit Project Details", along with submission status and due date information.

4.2 EVALUATOR UI

Evaluator Main Page

After Evaluator login successful, evaluator will see the session details

The screenshot shows a web application window titled "Evaluator Dashboard - Dr. Tan". The header area includes a welcome message "Welcome, Evaluator Dr. Tan" and a "Logout →" button. Below the header, there is a prompt "Please select a session to evaluate:". A table lists available sessions with columns for Session ID, Presentation Date & Time, Due Date & Time, Venue, and Students. At the bottom of the dashboard is an "Enter Session →" button.

Session ID	Presentation Date & Time	Due Date & Time	Venue	Students
SES-01	2026-02-10 10:00 AM	2026-02-10 11:59 PM	Lab 1	2 / 5
SES-02	2026-01-12 02:00 PM	2026-01-12 11:59 PM	Hall A	0 / 5

Evaluator Session Page

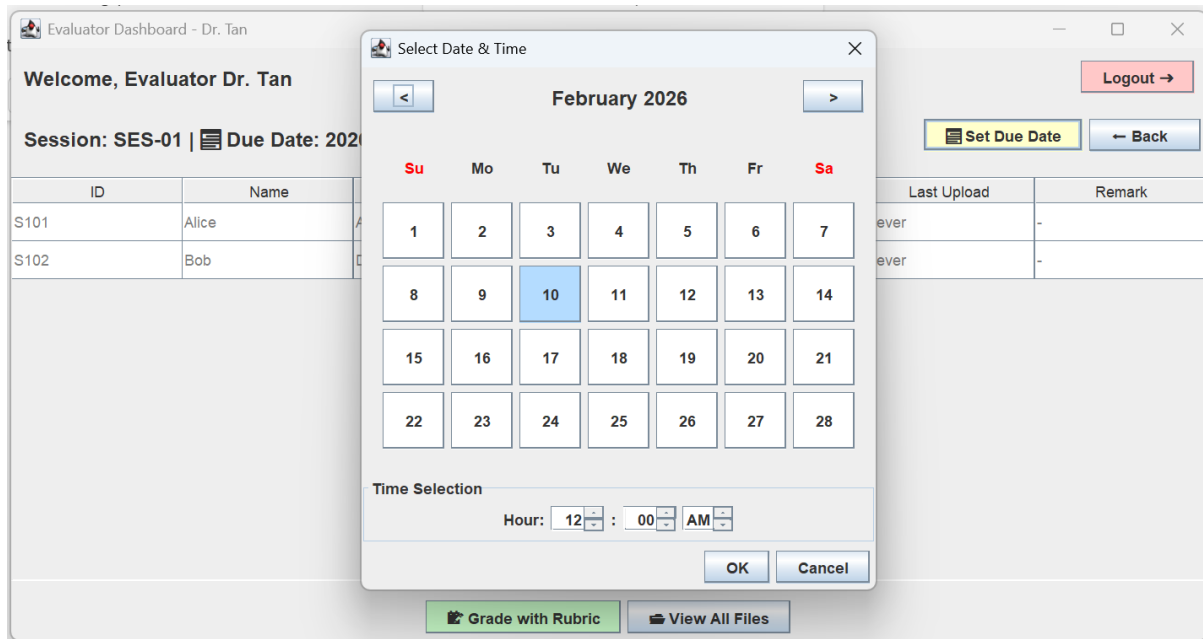
- Double Click session row, evaluator can view the students by session assigned by Coordinator.
- Evaluator able to view Student Submission Details, Grading and give some remarks.

The screenshot shows the "Evaluator Session Page" for Session SES-01. The header area includes a welcome message "Welcome, Evaluator Dr. Tan" and a "Logout →" button. Below the header, the session details are displayed: "Session: SES-01 | Due Date: 2026-02-10 11:59 PM". There are two buttons: "Set Due Date" and "Back". A table lists student submissions with columns for ID, Name, Research Title, Status, Score, Last Upload, and Remark. At the bottom of the page are two buttons: "Grade with Rubric" and "View All Files".

ID	Name	Research Title	Status	Score	Last Upload	Remark
S101	Alice	AI Research	Pending	-	Never	-
S102	Bob	Data Science	Pending	-	Never	-

Evaluator Select Session Due Date and Time

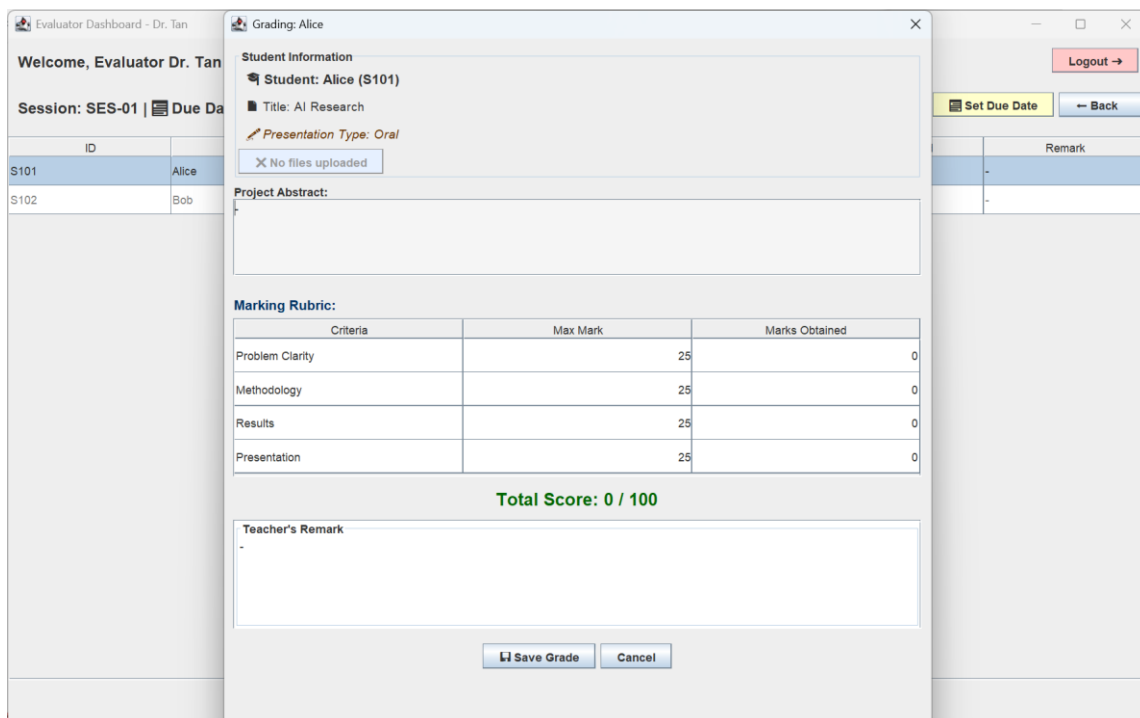
Evaluator also able to select the Due Date and Time Submission by Session.



Grade with Marking Rubric and Remark

Evaluator able to view the latest Submission File by student in Grading page.

After viewing, evaluator can directly to insert the Mark and Remark to save the Marking Details.



View All Files uploaded by Students

Evaluator also can view all the history file uploaded by students in “View All Files” button.

The screenshot shows the 'Evaluator Dashboard - Dr. Tan' interface. At the top, it says 'Welcome, Evaluator Dr. Tan' with a 'Logout →' button. Below this, it shows 'Session: SES-01 | Due Date: 2026-02-10 11:59 PM' with 'Set Due Date' and 'Back' buttons. A table lists student uploads:

ID	Name	Research Title	Status	Score	Last Upload	Remark
S101	Alice	AI Research	Pending	-	Never	-
S102	Bob	Data Science	Pending	-	Never	-

A modal window titled 'Files: Alice' is open, showing columns for 'File Name' and 'Upload Time'. At the bottom of the modal is an 'Open Selected File' button. At the bottom of the dashboard are 'Grade with Rubric' and 'View All Files' buttons.

4.3 COORDINATOR UI

Coordinator Main Page

Coordinators can directly view the Session Details, Best Award, and Unassigned Students on the main page. This design can let Coordinators view it straight to the point. It can avoid pressing multiple buttons to complete the task.

Welcome, Coordinator Dr. Admin Logout →

Presentation Schedules (Double-click row to view Students)

Session ID	Date	Time	Venue	Session Type	Evaluator	Availability	Edit Session
SES-01	2026-02-10	10:00 AM	Lab 1	Oral	Dr. Tan (E001)	2 / 5	Edit
SES-02	2026-01-12	02:00 PM	Hall A	Poster	Dr. Tan (E001)	0 / 5	Edit

Best Award Oral

Rank	Name	Session	Score
1	Alice	SES-01	-
2	Bob	SES-01	-

Best Award Poster

Rank	Name	Session	Score
1	Charlie	Unassigned	-

Unassigned Students (Direct Assign)

No	ID	Name	Title	Type	Assign Session
1	S103	Charlie	Cyber S...	Poster	- Select -

[+ Create Session](#) [Delete Session](#) [Add Evaluator Account](#) [Generate Reports](#) [Assign Session](#)

Create Session

Coordinator are able to Create Session.

Create Session ×

ID:

Date/Time:

Venue:

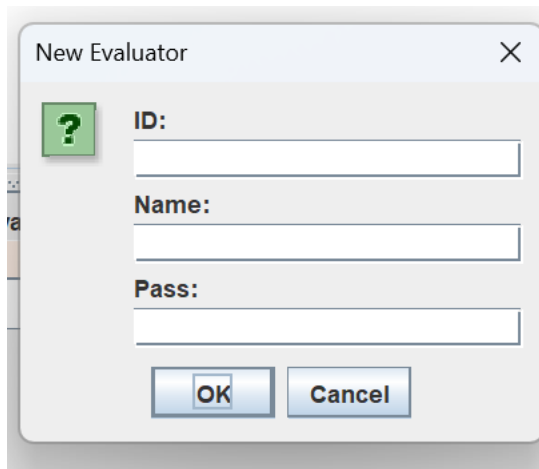
Cap:

Type:

[OK](#) [Cancel](#)

Add New Evaluator

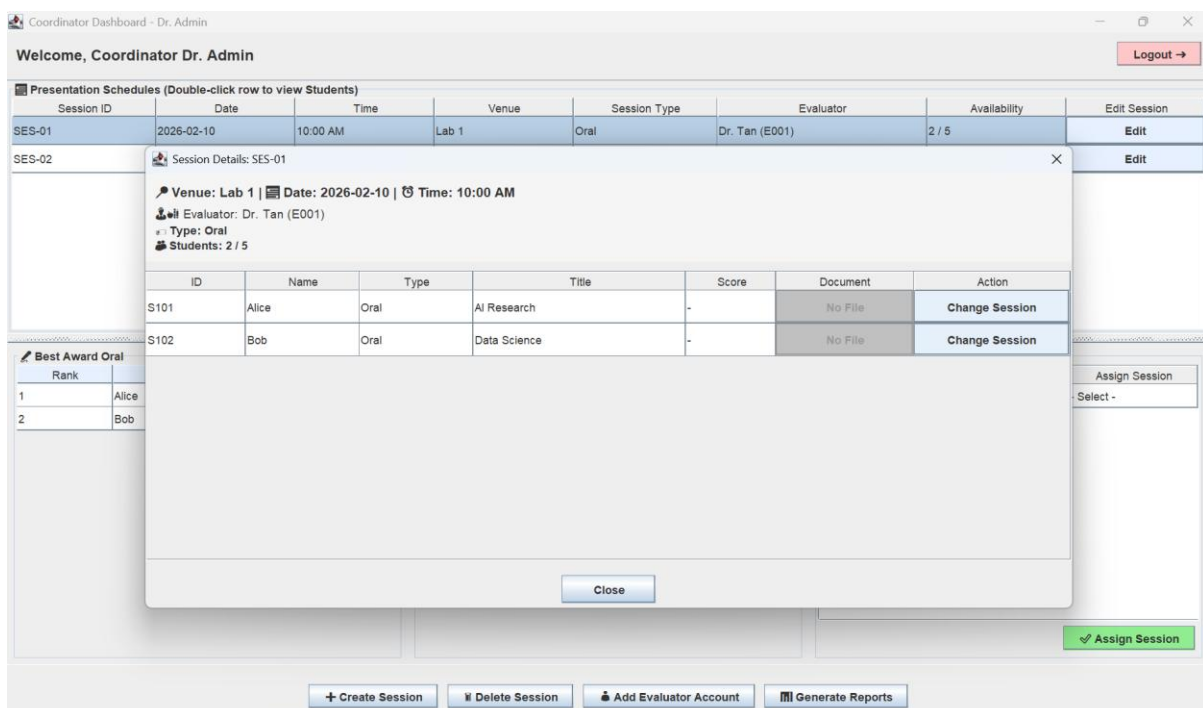
Evaluator Account can be created by Coordinator.



A dialog box titled "New Evaluator" with a close button (X) in the top right corner. It contains three input fields: "ID:" with a green question mark icon, "Name:", and "Pass:". Below the fields are two buttons: "OK" and "Cancel".

Session Details

Coordinator able to view students by session. Only Coordinator allow to change the student presentation session to the same Session Type. They are not allowed to change cross over the Session Type



The screenshot shows the "Coordinator Dashboard - Dr. Admin" interface. At the top, it says "Welcome, Coordinator Dr. Admin" with a "Logout" button. Below is a "Presentation Schedules" table with columns: Session ID, Date, Time, Venue, Session Type, Evaluator, Availability, and Edit Session. The table lists two sessions: SES-01 and SES-02. A modal window titled "Session Details: SES-01" is open, showing session information: Venue: Lab 1, Date: 2026-02-10, Time: 10:00 AM, Evaluator: Dr. Tan (E001), Type: Oral, and Students: 2 / 5. Below this is a table of students for the session:

ID	Name	Type	Title	Score	Document	Action
S101	Alice	Oral	AI Research	-	No File	Change Session
S102	Bob	Oral	Data Science	-	No File	Change Session

At the bottom of the dashboard, there are buttons for "Create Session", "Delete Session", "Add Evaluator Account", and "Generate Reports". A "Best Award Oral" table on the left shows a ranking of students. A "Assign Session" dropdown menu is on the right, and an "Assign Session" button is at the bottom right.

Assign Session for Students

Coordinator can assign the session for students via right corner below the main page. It can view directly to avoid miss-assign the session for students.

No	ID	Name	Title	Type	Assign Session
1	S103	Charlie	Cyber S...	Poster	- Select - - Select - SES-01 (Oral) [2/5] SES-02 (Poster) [0/5]

Assign Session

Generate Report

There have 4 reports type for Coordinator

They are allow to export out as Excel and Text file following by the report.

Seminar Schedule Report: Text file

Evaluation Report: Excel File

Generate Text File (txt) by System

Generate Seminar Schedule Report

Output:

Report:

- Seminar Schedule
- Seminar Schedule
- Evaluation Report
- Award Report
- Analytics Summary

Preview Report

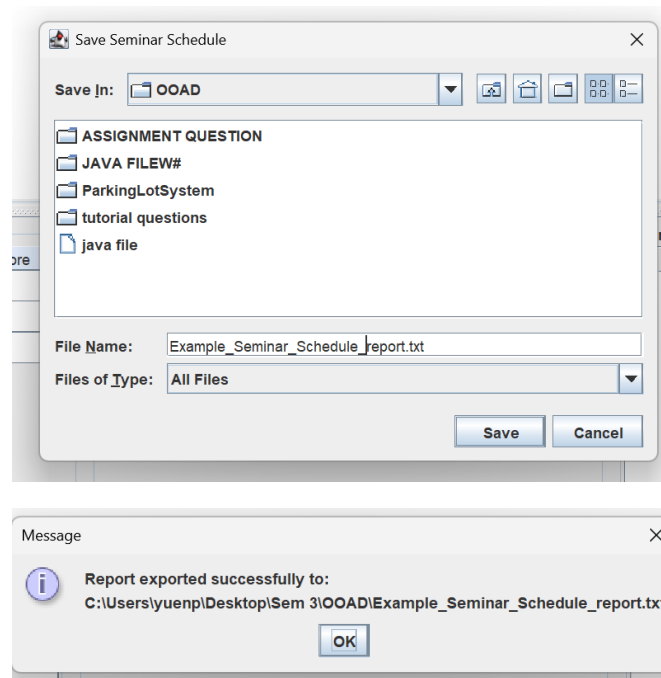
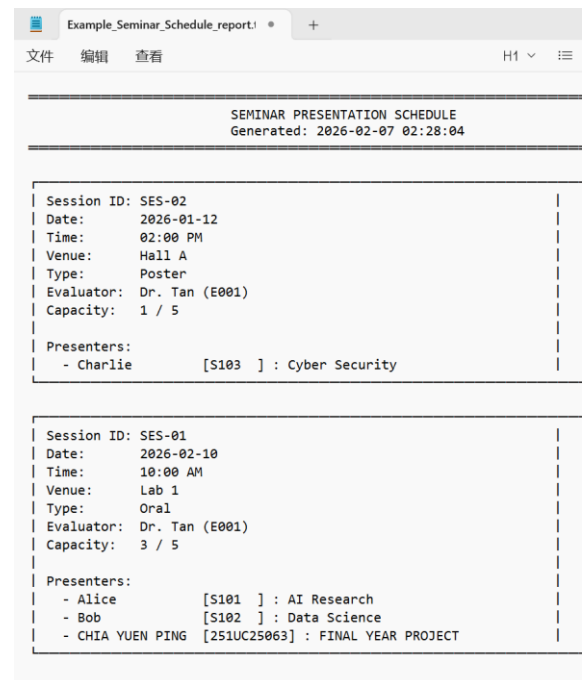
SEMINAR PRESENTATION SCHEDULE
Generated: 2026-02-07 02:28:04

Session ID: SES-02
Date: 2026-01-12
Time: 02:00 PM
Venue: Hall A
Type: Poster
Evaluator: Dr. Tan (B001)
Capacity: 1 / 5
Presenters:
- Charlie [S103] : Cyber Security

Session ID: SES-01
Date: 2026-02-10
Time: 10:00 AM
Venue: Lab 1
Type: Oral
Evaluator: Dr. Tan (B001)
Capacity: 3 / 5
Presenters:
- Alice [S101] : AI Research
- Bob [S102] : Data Science
- CHIA YUEN FING [251025063] : FINAL YEAR PROJECT

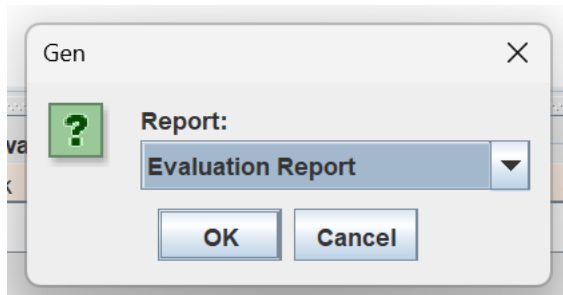
Do you want to export this report to file? (Click YES to Export)

Yes No

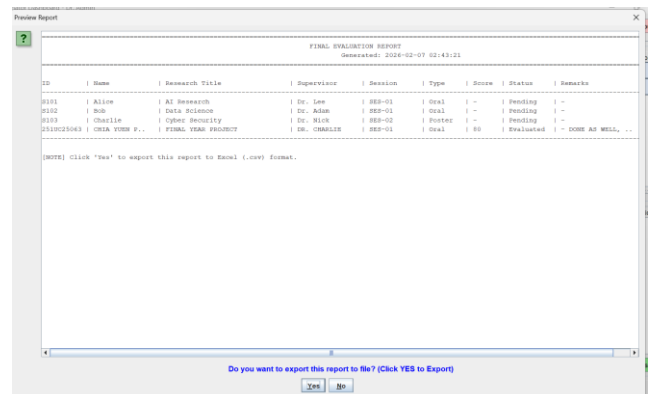
Save as TXT file**Export Text file:**

Generate Excel File (CSV) by System

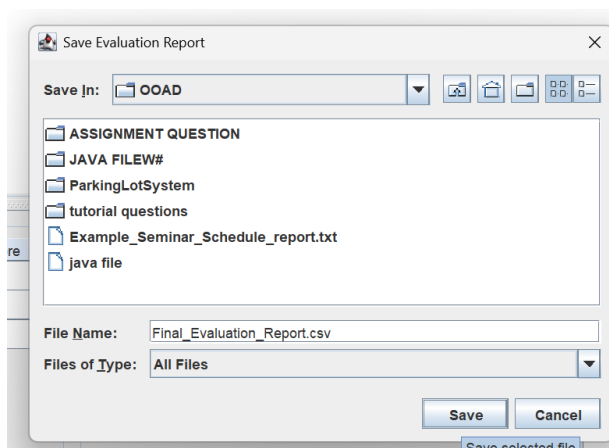
Generate Evaluation Report:



Evaluation Report:



Save Report as Excel file:



Export as Excel file:

